File Compression & Decompression System in C++

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Note: Slides (excluding Thank-you and Introduction slide) should not be more than 4.

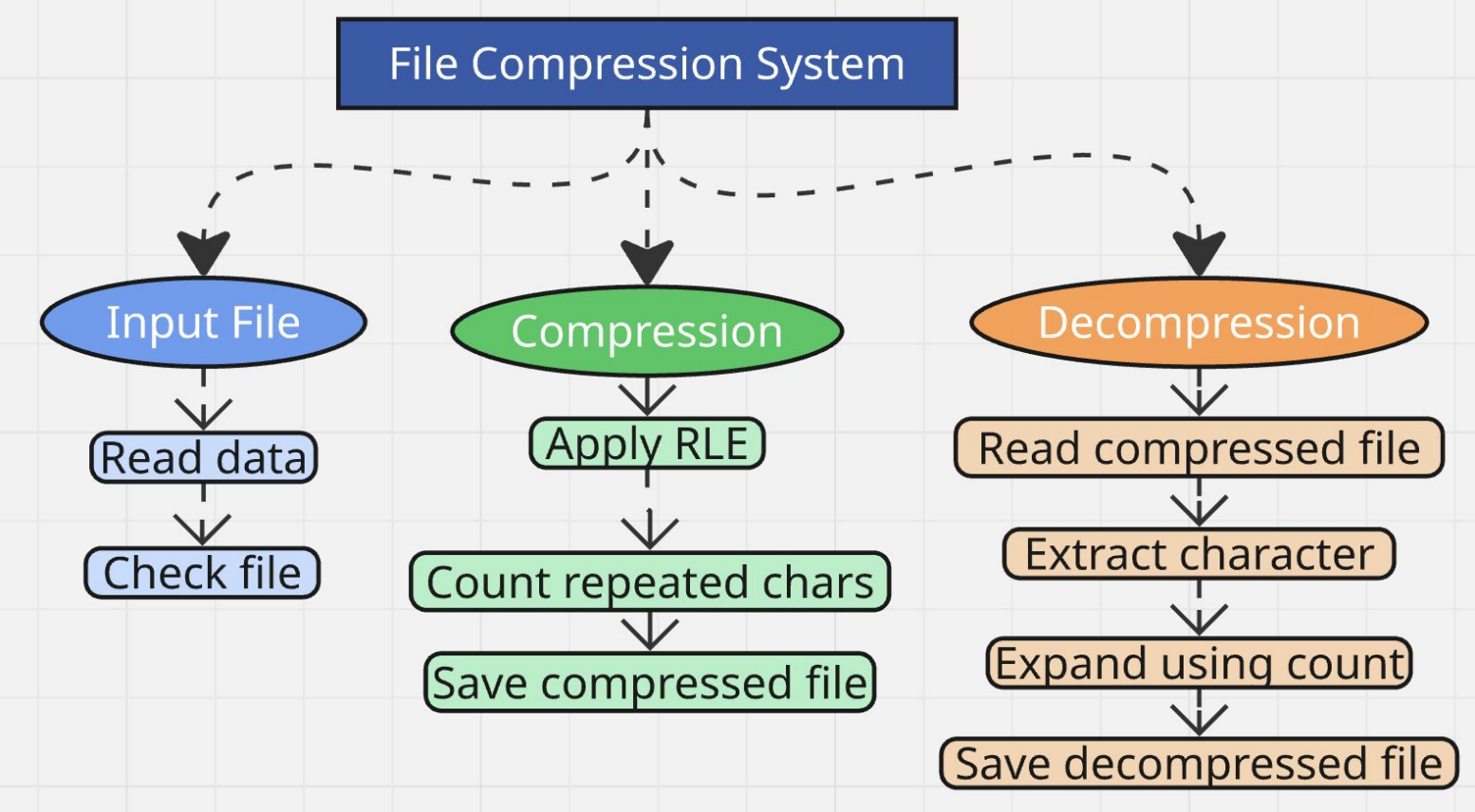
File Compression & Decompression System in C++

- Large files consume a lot of storage space.
- Efficient storage and fast file transfer are needed.
- Goal: Reduce file size without losing data.
- Users need a simple tool to compress and decompress files

Solution

- Developed a **C++ program** for file compression & decompression.
- Uses **Run-Length Encoding (RLE)**: replaces repeated characters with character + count.
- Example: aaaaa → a5
- Handles:
- File input/output
- Compression logic
- Decompression logic
- Provides **error handling** for empty or missing files.

Working (Method / Mind Map)



Conclusion

- File compression & decompression successfully implemented in C++.
- RLE technique effectively reduces repeated characters.
- Program handles basic file errors and saves output correctly.
- Can be extended for more advanced compression methods in the future.

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Thank You